Yeqi Huang

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Education

- **University of Edinburgh** Edinburgh, UK
  PhD in Computer Science *August 2023 - Present*
  Research Field: *AI-Systems, Distributed ML, ML-oriented Architecture, Serverless*
- **University of Science and Technology of China** Hefei, China
  Bachelor of Computer Science; The school of Gifted Young; *July 2017 - June 2021*
  Courses: *Operating Systems, Artificial Intelligence, Principles of Compiler, Computer Architecture, High Performance Architecture*

Publications

1 **Yeqi Huang**, Congjie He, Haocheng Xiao, et al. "Wavel: A Fast and Efficient Compilation System for Wafer-Scale Accelerators." *Accepted to the ACM Symposium on Operating Systems Principles (SOSP)*, 2026.

2 Congjie He, Le Xu, Zhan Lu, **Yeqi Huang**, et al. "MeshRT: Compile-Time Governed Wafer-Scale Runtime for Low-Latency High-Throughput Inference." *Accepted to the ACM Symposium on Operating Systems Principles (SOSP)*, 2026.

3 **Yeqi Huang**, Yanwei Ye, Guomin Chen, et al. "SwarmX: Agentic Cluster Scheduling Framework." *Under Review*, OSDI 2026.

4 **Yeqi Huang**, Yue Chen, Yanwei Ye, et al. "BioVLM: Evidence-Enriched Data Synthesis from Biomedical Papers." *Under Review*, ACL 2025 System Demonstration.

5 Chuanming Zha, Jiamin Zheng, **Yeqi Huang**, et al. "LLM-Monitor: Efficient Privacy Violation Monitoring for LLMs."

6 Yinsicheng Jiang*, **Yeqi Huang***, Liang Cheng, et al. "ContextPilot: Efficient Retrieval-Augmented Generation with Accuracy-Preserving Context Reuse." *MLSys*, 2026. (*Equal Contribution)

7 Congjie He, **Yeqi Huang**, Pei Mu, et al. "WaferLLM: Large Language Model Inference at Wafer Scale." *Under Review*, OSDI 2025.

8 Yao Fu*, Yinsicheng Jiang*, **Yeqi Huang***, et al. "MoE-CAP: Cost-Accuracy-Performance Benchmarking for Mixture-of-Experts Systems." *NeurIPS*, 2025. (*Equal Contribution)

9 Yao Fu, Leyang Xue, **Yeqi Huang**, et al. "ServerlessLLM: Locality-Enhanced Serverless Inference for Large Language Models." In *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24)*, 2024.

10 Xiao-Long Chen, Lin-Feng Wang, **Yeqi Huang**, et al. "Symplectic structure-preserving particle-in-cell whole-volume simulation of tokamak plasmas." In *SC21: International Conference for High Performance Computing, Networking, Storage and Analysis*, 2021.

Projects

- **QED Simulation on GPU**:
  - Implemented high-performance Monte-Carlo simulation achieving 1500x to 5000x speedup.
  - Received recognition from Nobel Laureate Frank Wilczek at APS conference.
  - Demonstrated expertise in CUDA programming and GPU micro-architecture optimization.
- **AutoReader Project**:
  - Built a vector database for the latest ArXiv papers for daily semantic search and subscription.(Full-stack AI application)
  - Achieving ***60× faster indexing than NVIDIA cuVS*** library and ***13× faster retrieval than Milvus***.
  - Extended system to support bioRxiv and PubMed papers with specialized biology-focused features.
  - Biology patched version is demostrating in Dartmouth university, will submit to Nature this year.

- **Wavel (SOSP 2026)**:
  - Designed a wafer-scale compilation system using MeshIR and the locality-preserving CUT and TUNE primitives.
  - Achieved up to 3.6× speedup over state-of-the-art compilers, 1.3–1.4× higher throughput than WaferLLM, and up to 2.97× over automatic baselines.
  - Matched or surpassed expert schedules on Cerebras WSE-3 and Tenstorrent Blackhole while reducing search from hours or weeks to minutes or seconds.
- **MeshRT (SOSP 2026)**:
  - Developed the first wafer-scale runtime to achieve both ultra-low latency and high throughput for LLM inference.
  - Compiled dynamic inference events into per-core schedules for decentralized execution with low runtime overhead.
  - Improved throughput by up to 30–68×, with 6–8× lower latency and 10–36× higher decode energy efficiency than H200 GPU inference engines in the low-latency regime.
- **SwarmX (submitted to OSDI 2026)**:
  - Presented the first scheduler agent framework that formulates cluster scheduling as an agentic intelligence problem.
  - Deployed in production on tens of thousands of servers, improving P99 latency by up to 50% and doubling throughput.
  - Designed modular agents with neural predictors and memory to optimize complex workflow performance.
  - Maintained compatibility with Ray and ComfyUI, enabling existing workflows to run without modification.
- **BioVLM (submitted to ACL 2025 System Demonstration)**:
  - Automated pipeline transforming biomedical papers into evidence-enriched VLM training data.
  - Progressive post-training (SFT + RL) on Qwen3-VL-8B, training BioVLM-8B at under $200 cost.
  - Achieved 48.0% weighted accuracy on LAB-Bench, 12.6% above base model and 3.8% above GPT-5.2.
- **LLM-Monitor**:
  - Designed a low-cost, high-quality, evolvable privacy violation monitoring system for LLMs.
  - Organized and compiled monitoring requirements into optimized representations executable by local medium-scale models.
  - Achieved 15× faster monitoring than GPT-5.2 using a specialized Qwen3-8B-based model.
  - Matched large public model (GPT-5.2) monitoring quality while being significantly faster and cheaper.
- **AI4Math - Sketchpad Project**:
  - Received grant from AI for Math Fund to develop Sketchpad system for formal mathematics (announcement)
  - Building system that automatically converts mathematical proofs into structured data using graph-based representation.
  - Enabling decomposition of proofs into individual statements for more precise auto-formalization.
- **More Projects (https://yeqi-huang.com/)**:
  - I have plenty of AI related projects on my personal page.

Talks

1 **Yeqi Huang**. ”InfiniTensor: A Tensor-Friendly, Efficient Parallel Programming Library for Accelerator-Centric Clusters.” *UKSys 2024*.

2 **Yeqi Huang**. ”Why we need a new clustering benchmark in AI retrieval?” *International Workshop on Efficient Generative AI*, 2024.

Teaching Experience

- **Machine Learning System TA (2025)**:
  - Teaching assistant for CUDA programming course, covering GPU architecture, CUDA, CuPy, and Triton.
  - Designed hands-on exercises and projects to help students master parallel programming concepts.
- **Machine Learning System TA (2026)**:
  - Teaching assistant with new CuTile programming module, teaching Tile-based GPU programming.

Awards

International Awards:

- International Supercomputing Conference Student Cluster Competition Champion - 2021
- Asian Supercomputer Conference Student Cluster Competition First Prize - 2021

National Awards:

- Best Chinese Supercomputing Application of the Year - 2022
- Huawei Pioneer Developer (4 in China) - 2021
- National Compiler Designing Competition Second Prize - 2021

Skills

- **AI-Related**:
  - In-depth understanding of LLMs, with experience porting high-performance inference and training frameworks to various hardware platforms, including Apple Silicon, GPUs, and Cerebras.
  - Strong understanding of vector retrieval, having developed and tested graph and vector databases on multiple platforms.
  - In-depth knowledge of Multi-Agent applications and developed efficient development components for such applications.
- **Computer Science Related**:
  - Highly skilled in **C++, Python, Rust**; familiar with Go, JavaScript, and Latex
  - Proficient with **CUDA, Intel ONEAPI, OpenMP** and related parallel and distributed programming
  - Well-versed in **LLVM**, frequently participating in LLVM Forum online discussions
  - Extensive experience working with Linux, including usage of eBPF
  - Rich experience in distributed systems and distributed machine learning frameworks
  - Strong engineering development experience, mastering various compiler-related tools and static analysis tools
- **Physics & Math**:
  - Highly skilled in Computational Fluid Dynamics and Molecular Dynamics
  - Strong knowledge in numerical methods and linear algebra
  - Proficient in Quantum Mechanics and Quantum Electrodynamics, with some knowledge of quantum computing

Special Experience

- **Open Source Enthusiast**: Contributing to notable projects like GNOME, LAMMPS, and LLVM, I've used GitHub since 2019 to showcase my development journey and ideas.
- **Running a Science-Themed Cafe**: Leveraging software development income, I established Quantum Coffee near my school—a collaborative space encouraging students to explore and discuss scientific topics across disciplines.
- **UNICEF Charity Projects**: I have donated 10% of all personal income to children's charities since 2019.